Cardiovascular Health — RAG Demonstration Guide

This document is an educational reference created for a Retrieval-Augmented Generation (RAG) demonstration project. It is not medical advice or a diagnostic guide.

1. What are cardiovascular diseases?

Cardiovascular diseases (CVDs) are a group of conditions affecting the heart and blood vessels. Examples include coronary heart disease, cerebrovascular disease, peripheral arterial disease, rheumatic heart disease, congenital heart disease, and conditions involving blood clots such as deep vein thrombosis and pulmonary embolism.

2. Common risk factors

- Tobacco use
- Unhealthy diet, including excessive salt, sugar, or unhealthy fats
- Physical inactivity
- Harmful use of alcohol
- High blood pressure
- High blood glucose
- Abnormal blood lipids such as high cholesterol
- Overweight and obesity
- Air pollution and other environmental factors

3. Prevention and risk reduction

Risk can often be reduced through healthy lifestyle choices and appropriate medical care. Common prevention measures include avoiding tobacco, eating a balanced diet, being physically active, maintaining a healthy weight, limiting harmful alcohol use, and getting appropriate care for conditions such as hypertension, diabetes, and abnormal blood lipids.

4. Common warning symptoms

Possible symptoms of a heart attack can include discomfort or pain in the centre of the chest and discomfort that may involve the arms, shoulder, jaw, or back. Shortness of breath, nausea, light-headedness, faintness, or cold sweating can also occur. Symptoms can vary, and urgent medical evaluation is important when a heart attack is suspected.

5. Why early detection matters

Some cardiovascular conditions may have few or no obvious symptoms before an acute event. Early identification of risk factors can support appropriate prevention, counselling, and medical management.

6. Use in this RAG project

The document is intended to demonstrate document ingestion, PDF text extraction, chunking, embedding generation, vector retrieval, and question answering. Example questions include: What are common

cardiovascular risk factors? What lifestyle factors can increase risk? What symptoms may occur during a heart attack?

Source for the factual background: World Health Organization, Cardiovascular diseases (CVDs), updated 31 July 2025. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-%28cvds%29